



» machinery, was carried out by gangs known as “Luddites.” The British government sent troops to control them, and at a mass trial in 1813, more than 50 people were sentenced to death or penal transportation to Australia (see pp.350–51), where they would be forced into unpaid labor.

Continental challenge

The first phase of the Industrial Revolution took place largely in Britain, but not exclusively so: Britain’s

◁ Eiffel Tower

Finished in 1889, Gustave Eiffel’s tower, made from 18,038 separate pieces of iron, was a sign of France’s industrialization. The tower was the world’s tallest structure for 41 years.

American colonies had shipbuilding and iron-production industries; and some German states began to industrialize their metalworking. At the start of the period, France’s total industrial output was close to Britain’s, but progress was stalled by the French Revolution (see pp.300–03). The first Continental nation to industrialize was Belgium, from 1820 onward.

A second wave, sometimes called the Second Industrial Revolution, took place in Germany, Switzerland, and the US after the development of their

railroads from 1840 to 1870. German industrialization accelerated massively after unification in 1871 (see pp.333), and by the turn of the century, both Germany and the US had overtaken Britain’s industrial output.

European industrialization was made possible not just by technology, but also by the availability of a workforce recently freed from serfdom—those

“**Avarice**, the spur of industry.”

DAVID HUME, SCOTTISH PHILOSOPHER, “OF CIVIL LIBERTY,” 1742



△ French railroad construction

As with industrialization itself, railroad building in France suffered a false start. France’s first railroad opened as early as 1832, but subsequent development was hindered by political and financial problems.

▷ Trading fuel

London’s Coal Exchange (1849) was symbolic of the Industrial Revolution in two ways: the existence of the exchange was evidence of the increased value of coal, while the building itself, with its 72-ft- (22-m-) iron and glass dome, would not previously have been possible.



SPREAD OF THE INDUSTRIAL REVOLUTION

From Britain, industrialization with its mechanization, market economy, and profound social change swept in waves across Europe, the United States, and Japan.

